IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 5 October 2025, accepted 20 October 2025, date of publication 30 October 2025, date of current version 5 November 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3627191

RESEARCH ARTICLE

Smart Metering Enabled by the Internet of Things for Monitoring Residential Energy Consumption and Power Quality

THALLYS B. DE SOUSA[1] AND RICARDO DE A. L. RABELO[2]
[1]Department of Electrical Engineering, Federal University of Piauí (UFPI), Teresina 64049-550, Brazil
[2]Department of Computer Science, UFPI, Teresina 64049-550, Brazil

Corresponding author: Thallys B. de Sousa (thallys_sccp@ufpi.edu.br)

This work was supported by the National Council for Scientific and Technological Development (CNPq) through the "Strategic Solutions for Aerial-Terrestrial Internet of Things (IoT) Network Architectures" under Grant 407274/2021-9 and through the "Development of an Intelligent Architecture for Applications in the Internet of Robotic Things" under Grant 307967/2022-0.

ABSTRACT This work presents the development of an Internet of Things based smart metering system designed for real-time detection and classification of electrical disturbances, with a focus on Power Quality monitoring and energy consumption analysis. The system is built around an ESP32 microcontroller, integrated with ZMPT101B voltage and SCT013 current sensors, to capture high-resolution electrical signals. To detect and classify anomalies such as voltage sags, swells, harmonic distortions, spikes, notches, and transients, the system applies advanced signal processing techniques, including root mean square computation and Fast Fourier Transform. The detection methodology complies with the Brazilian PRODIST standard, which defines reference thresholds for Power Quality parameters using the per-unit system. A physical prototype was constructed using 3D printing, and system behavior was validated through simulations conducted in Proteus. The prototype demonstrated reliable energy monitoring with data acquisition at ten-second intervals and achieved classification accuracy ranging from 87.1% to 94.4% across different types of electrical disturbances. By combining Power Quality analysis with consumption tracking, the proposed Internet of Things enabled smart meter provides a cost-effective and scalable solution for continuous monitoring. Its implementation supports improved efficiency and reliability in electrical systems and is well-suited for residential applications, enabling enhanced energy management and predictive maintenance through detailed consumption insights and disturbance detection.

INDEX TERMS Energy consumption, Internet of Things, power disturbances, power quality, smart meter.

I. INTRODUCTION

In recent years, technological advancements have accelerated, particularly in the realm of the Internet of Things (IoT), which connects devices in a network that continuously gathers and exchanges data [1]. This technology has found applications in multiple domains, including energy, healthcare, agriculture, transportation, industry, and smart cities [2]. Figure 1 showcases various IoT application areas, each highlighting the technology's potential to transform and optimize processes, boost efficiency, cut costs, and enhance quality of life.

Among these fields, energy management has emerged as a key area of opportunity. The increasing global energy demand, coupled with the need for reliability, availability, and security, highlights the urgency of implementing intelligent solutions for monitoring and optimizing electrical infrastructure [3].

Traditionally, electromechanical energy meters have been the standard for tracking electricity consumption in residential and commercial settings. These conventional meters, shown in Figure 2, register total energy usage over a specific

The associate editor coordinating the review of this manuscript and approving it for publication was Diego Bellan.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

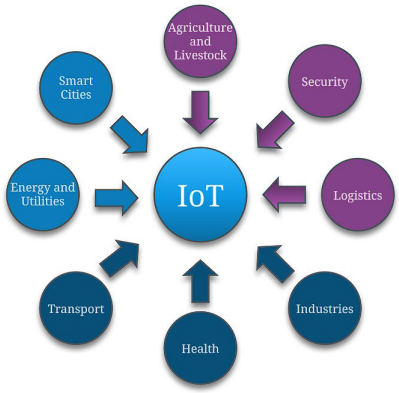


FIGURE 1. IoT application areas across various sectors.

period and present the data on an analog or digital display [4], [5].



FIGURE 2. Conventional single-phase electric consumption meter.

A conventional energy meter operates using current and voltage coils to generate magnetic fields proportional to energy consumption, causing an aluminum disc to rotate at a speed corresponding to the consumed energy [6]. However, these meters provide limited insight into the consumption of individual devices and do not capture Power Quality (PQ) parameters, such as electrical disturbances [7].

Figure 3 illustrates various electrical disturbances examined in this study, including voltage sags, swells, oscillatory transients, harmonics, spikes, and notches. These phenomena affect PQ and can lead to equipment damage, performance degradation, or system failures.

Voltage sags are temporary but significant reductions in electrical voltage within a power system. These events are often caused by the connection of heavy loads to the grid or by momentary system faults [9]. Conversely, voltage swells represent the opposite phenomenon, characterized by a brief yet substantial increase in voltage magnitude. This typically occurs due to a sudden reduction in the load connected to the electrical grid [10].

Oscillatory transients refer to rapid fluctuations in voltage or current that appear briefly, often triggered by load switching or operational maneuvers within the electrical system [11]. Harmonics, on the other hand, are frequency

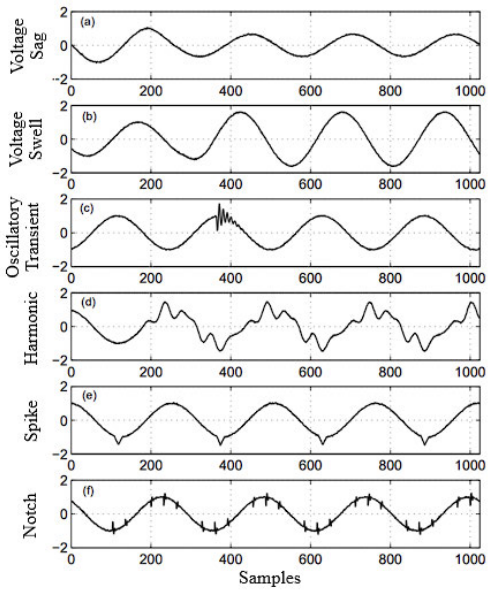


FIGURE 3. Some types of electrical disturbances, including voltage sag, voltage swell, oscillatory transient, harmonic, spike, and notch, over 1000 samples [8].

components that are integer multiples of the fundamental frequency in power systems. These are primarily generated by nonlinear electronic devices, such as inverters, computers, and solid-state loads, where voltage and current do not follow a linear relationship [12].

Spike disturbances, also known as surges, involve sudden and transient voltage or current variations that exceed normal system values. These events are typically caused by factors such as lightning strikes, switching operations, or electrical faults [13]. Notches, meanwhile, manifest as small interruptions or distortions in voltage or current waveforms, often resulting from nonlinear loads that introduce imbalances into the electrical system [14].

The increasing demand for electrical energy, driven by rapid technological advancements, has underscored the need for efficient strategies in the management and protection of power distribution networks. In this context, the IoT has emerged as a key enabler, offering intelligent solutions for real-time monitoring and efficient control of electrical infrastructure. The integration of IoT technologies has demonstrated significant potential in optimizing energy usage and enhancing the reliability of power systems.

Modern power grids face critical challenges, including load balancing, fault detection, and protection against electrical disturbances, which can lead to service interruptions, financial losses, and public safety concerns. Therefore, the development and deployment of robust, automated monitoring and protection systems have become essential to ensure grid stability and resilience. These systems must be capable of promptly identifying anomalies and responding effectively to mitigate risks and maintain uninterrupted power delivery.

This study presents a system designed to provide residential consumers with detailed insights into the energy consumption patterns of individual electrical appliances. Using the Blynk IoT platform, the system enables both the storage and remote access of consumption data, allowing users to monitor historical usage with high granularity. The real-time monitoring functionality supports data-driven decision-making, promoting more efficient energy use and potential cost reduction.

In addition to consumption monitoring, the proposed system incorporates PQ analysis by detecting and identifying electrical disturbances within the power network. These disturbances, including voltage sags, swells, harmonic distortions, and transient events, are accurately captured and displayed both on a local interface and within the Blynk IoT application. By providing immediate feedback on PQ anomalies, the system helps users identify conditions that may affect appliance performance and take appropriate corrective actions. This combination of energy monitoring and PQ analysis offers a complete solution for improving residential energy management and maintaining the reliability of electrical systems.

II. CONTEXT AND RELATED WORK

Humanity continually strives to develop and enhance innovative technologies that simplify daily life, with the IoT representing a major advancement in this important direction [15]. The detailed studies in [16] and [17] investigate significant energy savings achieved through continuous electrical consumption monitoring using advanced smart meters; however, they do not address specific Power Quality issues.

The research in [18] employs the ESP32 for residential energy monitoring but lacks details on electrical disturbances and does not incorporate a dedicated power supply for the prototype. Similarly, the study in [19] does not include a simulation program for model validation prior to implementation, nor does it provide users with access to additional electrical parameters such as voltage, current, and power.

Meanwhile, the studies in [20], [21], and [22] focus on real-time monitoring of electrical consumption. However, they do not offer a cloud-based platform for users to visualize the data. This article addresses that gap by leveraging the Blynk IoT platform as a key solution.

Table 1 summarizes key aspects of the cited studies, comparing methods and performance. It highlights this research's contributions, addressing gaps and improving solutions, emphasizing the system's novelty and relevance.

The developed prototype in this research incorporates essential functionalities for energy quality monitoring. It features an individualized smart meter designed to detect electrical disturbances and record critical variables. A dedicated power supply ensures operational autonomy. Model validation was conducted using simulation software, initially employing Proteus, a widely recognized tool for simulating microcontrollers, circuits, and Printed Circuit Boards (PCB) in embedded systems development [23].

TABLE 1. Comparison of key characteristics addressed by various authors in related studies.

Authors	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
[15]	×	-	-	×	-	-	-	×	×
[16]	×	-	-	×	-	-	-	×	-
[17]	-	-	-	-	-	-	-	×	-
[18]	×	×	-	×	-	-	-	×	×
[19]	-	×	-	-	-	-	-	-	×
[20]	×	-	-	-	-	-	-	×	-
[21]	×	-	-	-	-	-	-	×	-
[22]	×	-	-	-	×	×	-	×	-
Us Proposed	×	×	×	×	×	×	×	×	×

Notes: Individual smart meter(1). Own power supply(2). Simulation in software(3). Recording of other electrical variables(4). Detection of electrical disturbances(5). Mechanisms for load activation(6). Alert via sound warning(7). Low-cost system(8). Integration with IoT platforms(9).

Additionally, the system integrates load activation mechanisms, enhancing control over energy consumption. To alert users, an audible alarm system has been implemented, augmenting system interactivity. Noteworthy emphasis has been placed on accessibility, resulting in a cost-effective prototype. Furthermore, integration with IoT platforms facilitates remote monitoring and real-time analysis of collected data.

National and international organizations have increasingly focused on establishing standards for electrical PQ. In Brazil, the National Electric Energy Agency has led efforts to regulate quality across transmission and distribution systems through the PRODIST (Procedures for Electric Power Distribution), is part of the Brazilian electrical energy standards, in effect since 2008 [24].

PRODIST defines performance indicators and service quality standards to ensure reliable and efficient energy delivery. Its implementation enhances the safety and effectiveness of the electrical system, benefiting both consumers and utility providers [25]. Additionally, aligning these regulations with international standards supports interoperability and facilitates global energy exchange.

One of the primary goals of a power distribution system is to ensure continuous and reliable energy delivery to consumers. Although utility companies carefully plan their operations to meet this objective, distribution systems remain vulnerable to unforeseen events that can negatively impact PQ [26].

The growing reliance on computerized and electronically sensitive equipment has increased the demand for stricter PQ standards [27]. In parallel, the widespread use of nonlinear loads—such as microprocessor-based systems and switched-mode power supplies—has significantly affected residential,

commercial, and industrial electrical installations. This shift has led to a rise in power disturbances and a noticeable decline in overall PQ [28].

Nonlinear loads draw current in an intermittent and non-sinusoidal manner, introducing waveform distortions, harmonic content, and other phenomena that degrade PQ [29]. These effects can cause equipment overheating, malfunction of sensitive devices, and, in more severe cases, system failures [30].

Short-Duration Voltage Variations (SDVV) represent significant oscillations in the amplitude of the root mean square (RMS) voltage value over a time interval of less than three minutes. Table 2, extracted from module 8 of PRODIST, summarizes relevant information about short-duration voltage variations, relating them to duration and amplitude. These events can result from various causes, including network overloads, switching operations, or failures in electrical equipment, and may adversely impact the operation of devices sensitive to PQ [12], while recent work on energy-storage monitoring highlights similar concerns for grid-level reliability [31].

TABLE 2. Classification of short-duration voltage variations.

Classification	Denomination	Duration of Variation
Momentary Voltage Variation	Momentary Voltage Interruption	≤ 3 seconds
	Momentary Voltage Sag	≥ 1 cycle and ≤ 3 seconds
	Momentary Voltage Swell	≥ 1 cycle and ≤ 3 seconds
Temporary Voltage Variation	Temporary Voltage Interruption	> 3 seconds and ≤ 3 minutes
	Temporary Voltage Sag	> 3 seconds and ≤ 3 minutes
	Temporary Voltage Swell	> 3 seconds and ≤ 3 minutes

Complementary research using machine-learning techniques has demonstrated improved classification of power-quality disturbances in noisy networks [32], further motivating adaptive algorithms for IoT smart meters.''

In addition, recent IEEE works highlight integrated IoT architectures for real-time power monitoring and predictive maintenance, underscoring the relevance of cloud-based PQ diagnostics [33], [34].

Equation 1, also derived from module 8 of PRODIST, describes the calculation of voltage variation indicators. V_e represents the event amplitude, V_{res} is the residual voltage of the SDVV event, and V_{ref} indicates the reference voltage.

$$V_e = \frac{V_{res}}{V_{ref}} \times 100 \tag{1}$$

Equations 2 and 3 describe the time and frequency characteristics of these events in the network. The parameter Δt_e represents the event duration in milliseconds, where t_i and t_f correspond to the initial and final instants of the SDVV event, respectively. Furthermore, in the other equation, f_e denotes the occurrence frequency of SDVV events, while n represents the total number of SDVV events recorded during the observation period.

$$\Delta t_e = t_f - t_i \tag{2}$$

$$f_e = n \tag{3}$$

While previous IoT-based smart meters have primarily focused on monitoring energy consumption [15], [16], [17], [18], [19], [20], [21], [22], few explicitly target the combined detection and classification of multiple PQ disturbances in residential networks. Our system goes beyond existing approaches by integrating low-cost hardware with real-time Fast Fourier Transform (FFT) based analysis and PRODIST-compliant thresholds, enabling detection of sags, swells, harmonics, spikes, notches, and oscillatory transients in a single platform. This dual focus on consumption and PQ diagnostics fills a critical gap in affordable, residential-scale monitoring solutions.

The novelty of this work lies not only in the integration of existing technologies. It also introduces a unified and cost-effective framework capable of simultaneously monitoring energy consumption and classifying PQ disturbances in real time. Unlike previous approaches, the proposed system achieves compliance with the PRODIST standard while employing low-complexity algorithms suitable for embedded devices, thus enabling practical deployment on resource-constrained hardware such as the ESP32. This dual functionality, combined with IoT-based remote monitoring and control, advances the scientific understanding of how IoT-enabled smart meters can evolve from simple data acquisition devices to intelligent, autonomous diagnostic agents within residential power systems.

III. SYSTEM DEVELOPMENT

The subsequent subsections present a comprehensive overview of the development process of the smart meter system outlined in this study.

A. IOT NETWORK

As illustrated in Figure 4, the IoT network architecture implemented in this study offers a comprehensive overview of the system, which is engineered to monitor electrical consumption while detecting and classifying disturbances within the electrical grid. The Blynk IoT platform functions as a cloud-based intermediary, facilitating seamless real-time communication between the ESP32 microcontroller and web or mobile applications, as well as enabling data storage for subsequent analysis. This setup allows users to access detailed consumption data and disturbance alerts remotely, promoting proactive management and timely interventions. Moreover, the integration of cloud services enhances system scalability and reliability, supporting continuous monitoring and efficient data handling.

As illustrated in the Figure 4, simulations of the proposed module were carried out to validate its performance and functionality. The following sections will present these

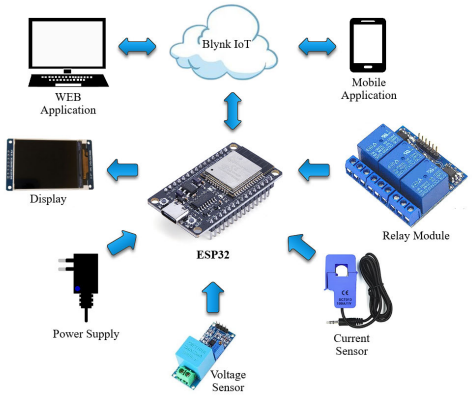


FIGURE 4. IoT network design.

simulations, demonstrating how the ESP32 microcontroller interacts with sensors, actuators, and the Blynk IoT platform, as well as its integration with Proteus.

In the proposed system, the ESP32 microcontroller establishes communication with the Blynk cloud server through a Wi-Fi connection, using the Message Queuing Telemetry Transport (MQTT) protocol. MQTT was selected due to its lightweight nature, low overhead, and proven suitability for IoT applications where bandwidth efficiency and fast message delivery are critical. This protocol enables bidirectional communication, allowing both the transmission of sensor data to the cloud and the reception of control commands from the user interface. By adopting MQTT over Wi-Fi, the system achieves reliable, scalable, and real-time communication between the residential monitoring device and remote users.

The integration of a robust communication system is essential in any IoT framework, as it ensures that local measurements are effectively synchronized with cloud-based services. In this work, Wi-Fi was chosen over alternative communication technologies (e.g., Zigbee, LoRa, or cellular) because of its availability in residential environments, ease of configuration, and compatibility with the Blynk IoT platform. This choice guarantees that the proposed system can be readily deployed in typical households without additional infrastructure. By explicitly addressing the communication layer, the system reinforces its practicality and highlights the central role of data transmission in enabling reliable monitoring and control of PQ and energy consumption.

B. SYSTEM DIAGRAM

Figure 5 presents a schematic diagram that consolidates all previously discussed components, offering a clearer understanding of the system architecture by illustrating their interconnections. The diagram includes the Direct Current (DC) power supply, the ESP32 microcontroller, voltage and current sensors, a display for result visualization, a relay module, and the connected electrical equipment. For system validation, three current sensors were connected to the electrical loads developed specifically for testing. This visual representation provides a comprehensive overview of the system layout, highlighting the functional relationships among components and their roles within the project.

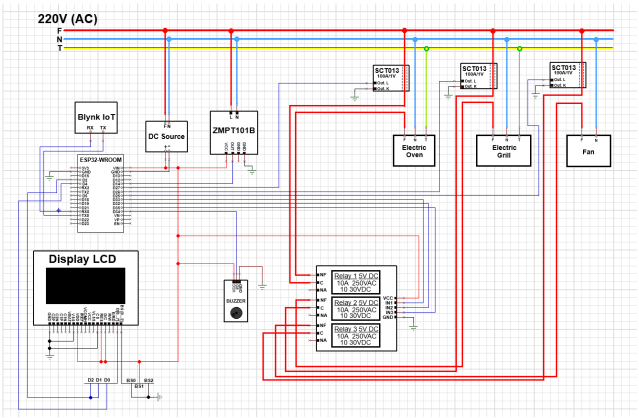


FIGURE 5. Diagram of the system developed for monitoring energy consumption and electrical disturbances in the power grid.

C. ESP32 PROGRAMMING

The first step in programming the ESP32 involves data acquisition from the sensors, specifically measuring mains voltage and electrical current when a load is in operation. This process is carried out by converting analog signals into digital ones using the ESP32's Analog-to-Digital Converter (ADC). As shown in Figure 6, which represents the steps related to the identification and detection of electrical disturbances.

The second step, signal processing, consists of two key stages: sampling and filtering. The primary objective is to extract meaningful information from the raw sensor data, as previously discussed, enabling the system to effectively detect electrical disturbances. To achieve this, the Fast Fourier Transform was applied to identify the phenomena mentioned in the Introduction.

Equation 4 represents the Discrete-Time Fourier Transform, commonly known as the Discrete Fourier Transform (DFT), where N denotes the total number of samples. The use of this equation is justified by the fact that the data collected during the acquisition stage consists of discrete samples of a continuous signal.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \tag{4}$$

The computation of the RMS value is carried out as demonstrated in Equation 5. This value is of utmost importance for obtaining a precise measurement of the amplitude of voltage and current signals over time. Moreover, it is essential for the identification of voltage swell and sag disturbances.

$$V_{RMS} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x[n]^2} \tag{5}$$

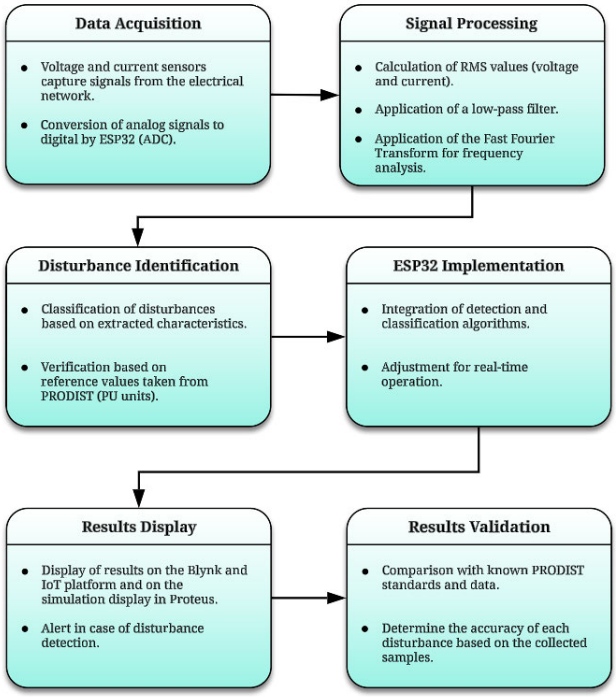


FIGURE 6. Flowchart representing the system for detecting and identifying electrical disturbances in the grid.

Prior to executing the RMS value calculation presented earlier, it was essential to filter the signals to minimize noise that could affect the analysis. To achieve this, Equation 6 was applied, representing a basic low-pass filter. This filter functions by smoothing the signals and isolating the relevant frequencies.

$$y[n] = a \cdot x[n] + (1 - a) \cdot y[n - 1] \tag{6}$$

The coefficient a in the low-pass filter was set to 0.1 after iterative experimental tests. This value was chosen to balance the trade-off between noise suppression and waveform fidelity. A smaller value of a resulted in excessive smoothing that compromised the accuracy of RMS and FFT analysis, while higher values allowed residual high-frequency noise to persist. The selected coefficient thus ensures effective noise reduction while preserving the essential characteristics of the measured signals.

After performing the FFT in the previous step, spikes and transients are identified by analyzing high-frequency components and sudden variations in the signal's magnitude. These phenomena can be detected by observing rapid and significant changes in amplitude within a short time frame.

For harmonics, the system needs to identify the presence of frequencies that are multiples of the fundamental frequency. To determine a harmonic of order n, the magnitude of the frequency component f_n is calculated as the product of the fundamental frequency (f_0, which in this case is 60 Hz) and the value of n. Equations 7 and 8 were used in the algorithm to identify spikes and harmonics, ensuring accurate analysis of electrical disturbances.

$$\text{Spikes} = x[n] - x[n - 1] > \Delta_{\text{spike}} \tag{7}$$

$$H_n = |X[n \times f_0]| \tag{8}$$

In the case of the notch disturbance, the procedure followed after the previously mentioned signal processing. The first step consists of detecting the zero-crossing, that is, identifying the points where the voltage waveform crosses zero. Next, the voltage derivative with respect to time is calculated, as described in Equation 9, allowing the detection of abrupt voltage variations within a short time interval.

$$\frac{dv}{dt} = \frac{V[n] - V[n - 1]}{T_s} \tag{9}$$

Additionally, it is essential to determine the threshold of this variation, as defined in Equation 10, as well as the depth of the notch, also expressed by Equation 11. For a complete characterization of the disturbance, it is necessary to measure the event duration, which corresponds to the time interval between the voltage drop and its recovery above the transfer threshold.

$$\left|\frac{dv}{dt}\right| > k \cdot \text{média}\left(\frac{dv}{dt}\right) \tag{10}$$

$$\Delta V = V_{\text{nominal}} - V_{\text{min}} \tag{11}$$

Finally, the following equations were employed to compute the active power, apparent power, power factor, and energy consumption of the electrical devices:

$$P = \frac{VI}{n_{\text{samples}}} \tag{12}$$

$$S = V_{\text{RMS}} \times I_{\text{rms}} \tag{13}$$

$$f_{\text{p}} = \frac{P}{S} \tag{14}$$

$$\text{Energy consumption (kWh)} = \frac{P \times t_{\text{ESP32}}}{3600 \times 10^3} \tag{15}$$

Equation 12 defines the power calculation based on the measured voltage V, current I, and the total number of samples acquired during the observation period. In relation to the energy consumption expressed in Equation 15, t_{ESP32} denotes the time interval registered by the ESP32 upon detecting an active load.

The electrical equipment's consumption is calculated considering the time elapsed since the ESP32 was initialized, using its internal timer. This approach ensures accurate measurement of energy usage over time, enabling precise tracking of consumption patterns and facilitating energy efficiency analysis. Additionally, the system can store and process the collected data for further analysis or transmission to external platforms for remote monitoring and control.

D. SIGNAL ACQUISITION AND PARAMETER CONFIGURATION

To ensure accurate detection of PQ disturbances on a low-cost microcontroller, all critical parameters were systematically calibrated through controlled laboratory experiments

employing programmable loads, a function generator, and an oscilloscope as a reference. The methodology was subsequently validated via pilot tests on a residential feeder.

- Sampling rate: Evaluated in the range of 2–8 kHz, with 4 kHz selected, enabling harmonic detection up to the 30th order while maintaining ESP32 Central Processing Unit (CPU) utilization below 10%.
- Windowing: A 2048-point Hanning window (1.95 Hz frequency resolution) was applied to reduce spectral leakage while limiting processing latency to under 150 ms.
- Overlap: A 50% overlap was adopted to improve temporal resolution for fast transients without causing memory overflow.

Regarding the characteristics of the filtering strategy parameters.

- Low-pass coefficient: Optimized to 0.12 by minimizing residual noise power while preserving step responses, achieving approximately -20 dB/dec attenuation.
- Anti-aliasing filter: An external RC filter (cutoff frequency $\approx$ 1.5 kHz) was implemented to suppress high-frequency noise prior to ADC conversion.
- Dynamic noise adaptation: The system measures baseline noise every 5 s and adjusts the filter coefficient by $\pm$10% when the signal-to-noise ratio (SNR) falls below 25 dB.

The RMS computation uses an integration window set to exactly one mains cycle (16.7 ms) and applies a rolling average over three cycles to minimize flicker in real-time displays.

Event-specific thresholds are defined as follows: spike detection is set to 1.8 times the noise standard deviation and updated every minute to track slow drift; the notch detection constant is 3.0, chosen to balance sensitivity and immunity to normal switching events; and harmonic order selection includes explicit calculation of harmonics up to the 25th order, with an aggregated THD metric reported for higher orders.

The ESP32 real-time clock is Network Time Protocol (NTP) synced via Wi-Fi every hour to keep the timestamp error under $\pm$50 ms. The firmware performs self-diagnostics, logging ADC saturation or buffer overruns to an internal register and triggering a Blynk alert if anomalies persist.

E. BLYNK IOT INTEGRATION

The integration of the Proteus simulation environment with the Blynk platform represents a significant advancement in this study. This integration enables real-time visualization and remote control of data acquired from sensors and electronic components within the simulated system, bridging the gap between virtual testing and practical application. By combining these tools, the system offers a flexible and scalable framework for validating device behavior and user interactions before physical implementation. This approach not only enhances the accuracy of the development process but also reduces time and costs associated with prototyping and troubleshooting.

Although similar approaches have been explored in previous works [18], the present implementation extends these efforts by leveraging the Blynk IoT platform to provide enhanced remote control features and more sophisticated data analysis capabilities. Users can access the platform via the cloud or the Blynk mobile application, allowing the creation of customized control dashboards for real-time device monitoring and interaction.

Figure 7 illustrates the data stream configuration within the Blynk IoT environment, detailing the monitored or controlled variables such as voltage, current, power, energy consumption, and relay states. Each data stream is defined by a unique identifier, label, alias, color code, virtual pin, data type, unit of measurement, raw data indicator, and preset minimum and maximum thresholds. These configurations ensure accurate, organized, and intuitive visualization of system data on the platform.

Id	Name	Alias	Color	Pin	Data Type	Units	Is Raw	Min	Max
5	Tensão	Tension		V0	Double	V	false	0	250
2	Corrente	Current		V1	Double	A	false	0	100
3	Potência	Power		V2	Double	W	false	0	2500
4	Consumo de energia 01	Energy Consumption 01		V3	Double	kWh	false	0	5
7	Consumo de energia 02	Energy Consumption 02		V4	Double	kWh	false	0	5
6	Consumo de energia 03	Energy Consumption 03		V5	Double	kWh	false	0	5
8	Módulo de Relé 01	Relay Module 01		V6	Integer		false	0	1
9	Módulo de Relé 02	Relay Module 02		V7	Integer		false	0	1
10	Módulo de Relé 03	Relay Module 03		V8	Integer		false	0	1
11	Módulo Buzzer	Buzzer Module		V9	Integer		false	0	1

FIGURE 7. Configurations and details of the Datastreams implemented in Blynk for control and visualization of the results.

F. COST ANALYSIS

As part of the feasibility assessment of the proposed system, a market survey was conducted to estimate the costs associated with the electrical components and modules described in this work. The prices were obtained through consultation of various e-commerce platforms, and the average values were subsequently calculated. Table 3 presents the list of all components, along with their respective quantities and average prices.

In the comparative analysis, it was found that the total cost of the developed system was lower than that reported in studies [15], [19], and [22], as discussed in Section II. Moreover, although the cost was comparable to that reported in studies [17], [20], and [21], the system presented herein incorporates additional functionalities pertinent to PQ analysis, particularly with respect to the detection of disturbances in the electrical grid.

TABLE 3. Budget for the components used in the system development.

Number	Description	Qty	Unit Price (US$)	Total Price (US$)
1	Microcontroller ESP32	1	30.00	30.00
2	Transformer 220V to 15V	1	23.00	23.00
3	Diode 1N4007	4	0.50	2.00
4	Electrolytic Capacitor 1000µF 25V	1	1.00	1.00
5	Electrolytic Capacitor 1µF 16V	1	0.50	0.50
6	Resistors (24 Ω, 10kΩ, 5kΩ)	1	2.00	2.00
7	Voltage Sensor ZMPT101B	3	10.00	30.00
8	Current Sensor SCT013 100A/1V	3	15.00	45.00
9	Buzzer Module	1	6.00	6.00
10	OLED Display	1	18.00	18.00

IV. SIMULATION AND IMPLEMENTATION OF THE PROTOTYPE

A full-wave bridge rectifier circuit was designed to provide a stable DC power supply for the ESP32 and associated components. This configuration converts the Alternating Current (AC) input into a DC output, ensuring reliable voltage regulation. Figure 8 illustrates the simulation of the rectifier circuit employed to power the ESP32. The output voltage measured immediately after the filtering capacitor and voltage regulator was 5.01 V, confirming the effectiveness of the filtering and regulation stages in maintaining the required voltage level.

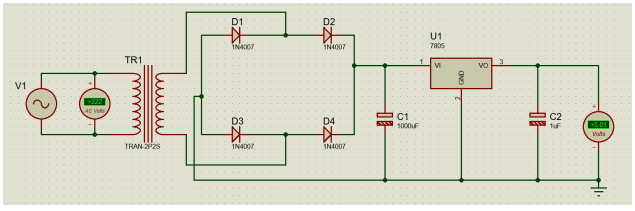


FIGURE 8. Power supply circuit developed for the prototype, including a full-wave rectifier and a filter for the DC signal.

Figure 9 depicts the test version of the power supply prototype. Designed to operate with a 220V AC input, the circuit delivers a regulated 5.02V DC output, as verified through multimeter measurements conducted during this study. The output stage includes a Universal Serial Bus (USB) Type-C connector for direct interfacing with the ESP32, as well as an additional DC output for powering other peripheral components. It is also important to note that the circuit was extensively tested on a protoboard to ensure functionality and reliability prior to its integration into the final PCB design.

The system illustrated in Figure 5, initially represented as a schematic diagram, was designed and simulated using Proteus. Its finalized implementation is presented in Figure 10. The setup performs continuous monitoring of the mains voltage through the ZMPT101B sensor and measures

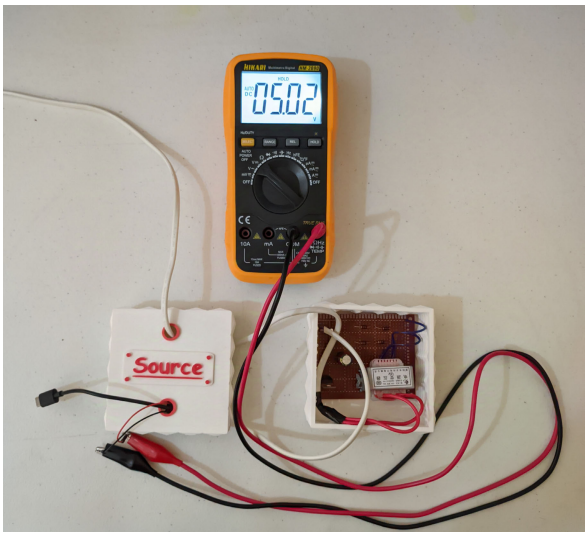


FIGURE 9. Verification of the voltage using a multimeter at the output of the designed power supply.

the electrical current of connected loads using the SCT013 sensor. The ESP32 microcontroller oversees the operation of electrical appliances via a relay module, enabling automated switching based on system conditions.

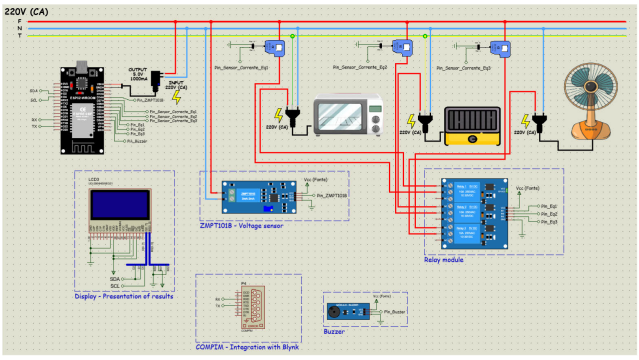


FIGURE 10. General simulation of the developed system performed in Proteus.

This functionality is particularly critical in scenarios involving electrical disturbances, as it allows for the automatic disconnection of devices, thereby mitigating the risk of damage due to voltage anomalies. In specific cases, such as overvoltage or undervoltage events, an audible alert is generated via a buzzer. Simultaneously, notifications are displayed on a local screen and transmitted to the Blynk IoT platform, ensuring timely remote alerts and enhanced system responsiveness.

Figure 11 displays the finalized version of the developed prototype. All mechanical components were designed using Fusion 360 and manufactured with a Creality CR-200B 3D printer. The enclosure houses the ESP32 microcontroller, the ZMPT101B voltage sensor, and the buzzer, in addition to a custom PCB that integrates all required electrical connections. To enhance system functionality, a relay module

was included to enable the disconnection of electrical loads in response to detected anomalies.

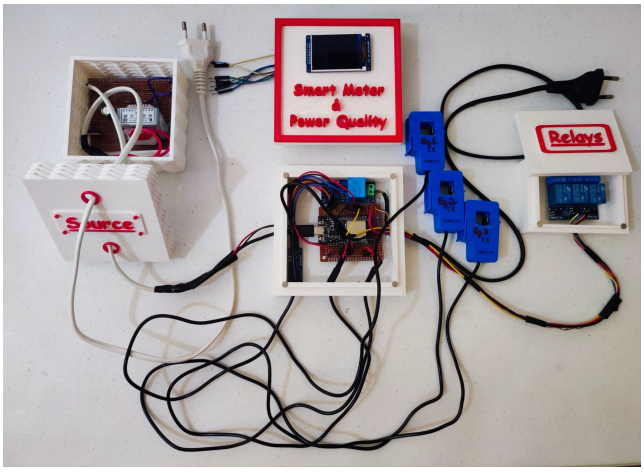


FIGURE 11. Overview of the prototype, highlighting the internal part of each compartment, its circuits, and respective connections.

The SCT013 sensors (100A/1V) were mounted externally to allow for convenient installation on a load-bearing conductor or the output of a conventional energy meter. These sensors facilitate real-time monitoring of household electrical consumption, provided the current remains within the 100-ampere operational limit of the SCT013.

V. RESULTS AND DISCUSSIONS

The prototype depicted in Figure 12 presents data related to the monitoring of electrical energy consumption for a fan (Eq3), categorized as an inductive load. During approximately 8 hours of continuous operation, the system recorded a total energy consumption of 2.72 kWh. This measurement is derived from the analysis of voltage, current, and power factor values, which are updated at 10-second intervals to enhance the precision of the readings. In scenarios where no additional loads (such as Eq1 or Eq2) are connected, the system displays only the voltage value measured by the ZMPT101B sensor. Furthermore, the prototype maintains a cumulative log of energy consumption, enabling the monitoring of total usage across extended periods of operation—for example, to evaluate the seasonal performance of specific appliances.

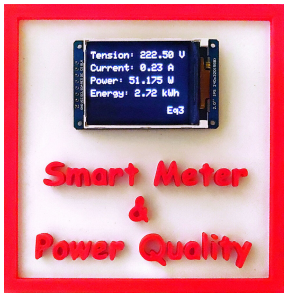


FIGURE 12. Display screen showing the energy consumption results of a fan during approximately 8 hours of operation.

The system is capable of detecting and identifying any of the six electrical disturbances illustrated in Figure 3. The corresponding alert messages displayed on the prototype are shown in Figure 13. Upon detection, the system notifies the user through multiple channels: the prototype's display, an audible alarm activated by a buzzer, and the Blynk IoT application, each specifying the nature of the detected disturbance.



FIGURE 13. Display screen indicating the detection of a disturbance in the electrical grid, such as a Voltage Sag.

This multi-modal alert system ensures prompt user awareness, facilitating timely intervention. Additionally, the platform provides configurable options for automatic equipment shutdown via relays, thereby enhancing protection against electrical anomalies and minimizing potential damage to connected devices.

The system's real-time monitoring and response capabilities contribute to improved PQ management and increased overall reliability of the electrical installation. Moreover, historical data on disturbances are logged within the IoT platform, enabling detailed analysis and predictive maintenance strategies.

Figure 14 displays the Web Dashboard configured on Blynk IoT, providing users with a convenient way to monitor all system-collected data remotely via the website or mobile app. The dashboard includes control buttons for operating a three-channel relay module and a text box that informs users about the presence and type of electrical disturbances detected in the network.

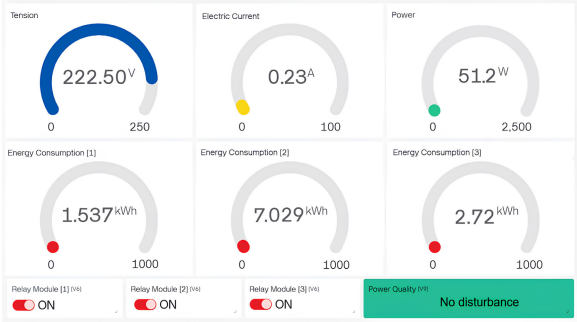


FIGURE 14. Web dashboard layout in Blynk IoT for monitoring electrical consumption and power quality.

The graphs shown in Figure 15 were generated on the Blynk IoT platform, offering a clearer visualization of the user's electrical consumption data. The graph on the left represents the consumption of an electric oven, which was operated intermittently for 15 minutes per day over a span of 7 days. The accumulated consumption is evident, as the system retains previous usage records. Meanwhile, the graph on the right illustrates the consumption of an electric grill, used daily for approximately 30 minutes during the same 7-day period.

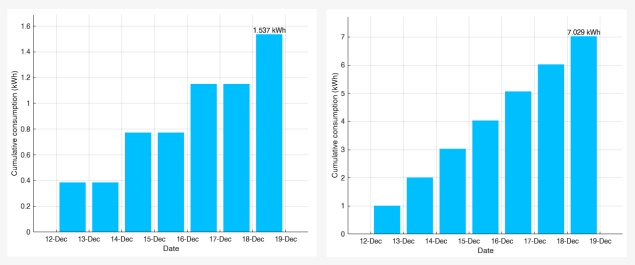


FIGURE 15. Cumulative electric energy consumption displayed on the Blynk IoT platform: on the left, the consumption of the electric oven; on the right, the consumption of the electric grill.

Figure 16 presents a comparison of the electrical consumption of a fan, operated for 8 hours daily over 7 days, with data obtained from a Hikari HM-2090 multimeter under three different conditions. The proposed system continuously measures voltage, current, and power every 10 seconds, ensuring precise calculation of electrical consumption, as demonstrated in the graph.

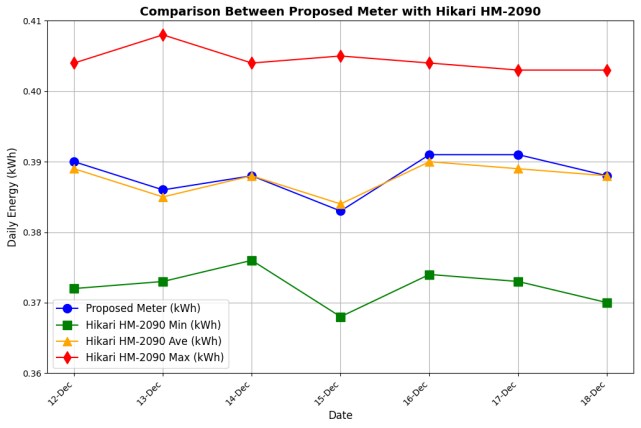


FIGURE 16. Comparison of the electric energy consumption of the fan based on three scenarios analyzed with the Hikari HM-2090 multimeter.

Figure 17 presents a comparison of the energy consumption of an electric oven and an electric grill with the average values recorded by the multimeter. The close agreement between the measurements indicates that the proposed system provides accurate monitoring of energy consumption for both appliances.

Table 4 presents a comparison between the values recorded by the Hikari HM-2090 multimeter and those measured by

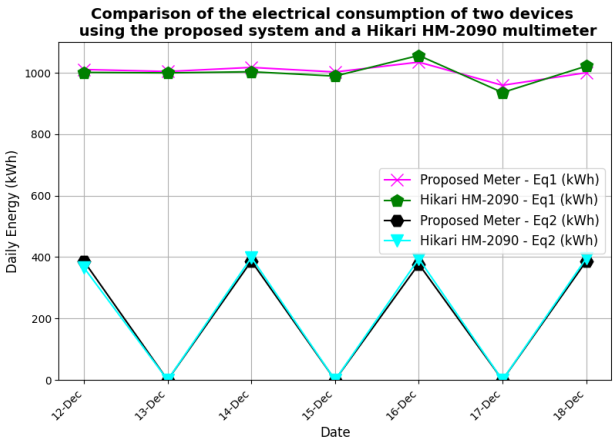


FIGURE 17. Comparative chart of the consumption of equipment Eq1 and Eq2 considering the results obtained by the developed system using the multimeter and its average consumption value.

the developed prototype over a 30-day monitoring period. The evaluated parameters—voltage, current, power, and energy consumption—showed a strong correlation between both measurement methods, indicating a high level of accuracy and consistency in the prototype's performance.

TABLE 4. Comparison between measurements recorded by the multimeter and the system for different devices over a 30-day period.

Quantities	Electric oven	Electric grill	Fan
Voltage recorded by multimeter (V)	220.30	220.32	220.42
Voltage recorded by system (V)	220.61	220.63	220.55
Current recorded by multimeter (A)	4.52	9.03	0.225
Current recorded by system (A)	4.50	9.00	0.210
Nominal power (W)	1000	2000	50
Power recorded by multimeter (W)	995.76	1989.5	49.60
Power recorded by system (W)	999.00	1999.00	49.00
Consumption recorded by multimeter (kWh)	11.9376	23.8752	5.9472
Consumption recorded by system (kWh)	11.9520	23.9040	5.7600
Consumption error (%)	0.1208	0.1207	3.25

Minor deviations observed fall within acceptable error margins, further validating the reliability of the proposed system. It is noteworthy that a slightly higher percentage error was detected in measurements associated with the electric fan, in comparison to other tested devices. This variance is likely due to the fan's non-purely resistive behavior, as its inductive or capacitive components can introduce fluctuations that affect the precision of the measurements.

In addition to the experiments performed with specific appliances, the proposed system can be extended to different residential use cases. For instance, in small households, the smart meter can track the consumption of basic appliances such as refrigerators, televisions, and lighting, enabling

families to identify unnecessary standby loads. In medium-sized apartments, where devices such as air conditioners, washing machines, and electric showers are common, the system provides detailed monitoring of high-demand appliances, supporting decisions to reduce peak consumption. In larger residences, where multiple high-power devices may operate simultaneously, the smart meter enables the identification of cumulative demand patterns and supports the prevention of overloading. These examples illustrate how the developed solution can adapt to diverse residential contexts, providing actionable insights for efficient energy management.

Finally, the proposed system was evaluated for its ability to detect and identify electrical disturbances over a 14-day period. To assess its accuracy, a Hantek 6022BE digital oscilloscope was used as a reference, with a total of 144 samples collected.

In this work, the term *real-time* refers to the capability of the system to both update energy consumption data and generate immediate alerts when a PQ disturbance is detected. Specifically, the prototype records and transmits consumption measurements at intervals of 10 seconds, enabling users to track energy usage with high granularity. In contrast, disturbance detection events are processed with sub-second latency, typically under 500 ms, ensuring that notifications are issued almost instantaneously through the local display, audible alarm, and Blynk IoT platform. This definition establishes a clear distinction between periodic consumption monitoring and immediate PQ event reporting, aligning the system's performance with the requirements of residential applications.

Figure 18 presents the accuracy results for different types of disturbances, demonstrating the system's effectiveness. The highest accuracy was observed for Voltage Sag, reaching 94.4% with 18 samples analyzed. Similarly, the system accurately identified Voltage Swell with 93.3% accuracy across 15 samples.

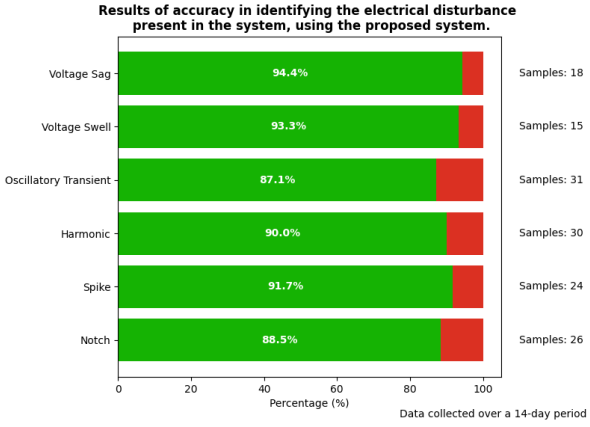


FIGURE 18. Comparison of accuracy rates in the identification of electrical disturbances by the proposed system, detailed by type of phenomenon and number of samples collected over a 14-day period.

To provide deeper insight into the classifier's performance, Table 5 presents the confusion matrix obtained during the 14-day monitoring period. This representation highlights not only the correctly identified disturbances but also the rare instances of misclassification between categories. For example, a small number of oscillatory transients were misidentified as spikes due to their similar high-frequency components. By illustrating both accuracy and error distributions, the confusion matrix complements the bar chart and enhances the interpretability of the system's performance in real-time PQ classification.

TABLE 5. Confusion matrix of PQ disturbance classification (14-day monitoring, 144 samples).

True / Pred.	Sag	Swell	Harmonic	Transient	Spike	Notch
Sag	17	1	0	0	0	0
Swell	1	14	0	0	0	0
Harmonic	0	0	27	1	1	1
Transient	0	0	1	27	2	1
Spike	0	0	0	1	22	1
Notch	0	0	1	1	1	23

Table 5 presents the confusion matrix obtained during the 14-day monitoring period, comprising a total of 144 samples. The highest classification accuracy was observed for voltage sags (94.4%) and swells (93.3%), with only one misclassification in each case. Harmonic disturbances were correctly identified in 90% of the cases, while notches achieved 88.5% accuracy. For transients and spikes, the system reached 87.1% and 91.7% accuracy, respectively, with occasional misclassifications occurring mainly between these two categories due to their similar high-frequency components. Overall, all disturbance types achieved accuracies above 87%, confirming the robustness of the proposed approach and demonstrating its effectiveness in real-time PQ disturbance detection and classification.

For experimental validation, a Hantek 6022BE digital oscilloscope was employed as the reference instrument. This oscilloscope features a sampling rate of 48 MSa/s, a bandwidth of 20 MHz, and an 8-bit vertical resolution per channel. These specifications ensure sufficient temporal and amplitude resolution for capturing and characterizing short-duration PQ events, providing a reliable benchmark for comparison with the proposed prototype.

For transient disturbances, the system achieved 87.1% accuracy in detecting Oscillatory Transients based on 31 samples, while Spike detection reached 91.7% accuracy with 24 samples. Harmonic disturbances were identified with 90.0% accuracy across 30 samples, and Notch disturbances were detected with an accuracy of 88.5%, considering 26 samples.

Overall, the proposed system demonstrated a high level of accuracy, consistently above 87% for all types of disturbances, confirming its reliability as a tool for electrical disturbance identification and classification.

As previously mentioned, the system's accuracy was evaluated based on the collection of signals by the oscilloscope at the moment the disturbance was detected by the system. For the use of the Hantek oscilloscope in the network, differential probes and a step-down transformer were employed, as the maximum voltage supported per channel by the oscilloscope is 35V. The acquired signals were processed and analyzed to verify the correspondence between the disturbances detected by the prototype and those recorded by the oscilloscope.

Figure 19 presents an example of a disturbance detected by the prototype, which was accurately identified as a voltage swell. The classification process was based on the criteria established in Table 2 of the PRODIST standard, which defines the permissible thresholds and characteristics for various types of electrical disturbances in distribution networks. The system continuously monitors the voltage waveform, and upon detecting a deviation, it compares the measured values against the standardized limits defined by PRODIST. In this case, the increase in voltage amplitude exceeded the upper threshold for the nominal voltage range, confirming the event as a voltage swell. This automated classification process ensures compliance with regulatory standards and enhances the system's ability to provide reliable PQ diagnostics in real time.

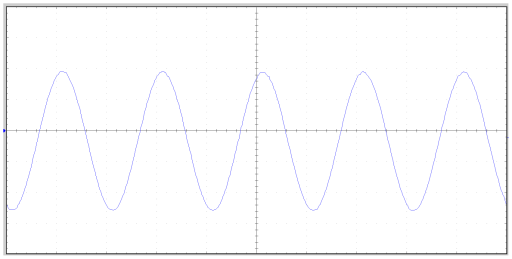


FIGURE 19. Voltage swell classified according to Brazilian energy standards from PRODIST.

Figure 20 presents a voltage sag event recorded by the Hantek oscilloscope. A momentary reduction in the amplitude of the sinusoidal waveform is observed, which is characteristic of this type of electrical disturbance. This phenomenon can be caused by the activation of high-power loads, short circuits, or other transient events in the electrical grid. The event was also identified by the developed prototype, following the PRODIST guidelines for characterizing voltage sags. The accurate detection of this disturbance demonstrates the proposed system's ability to monitor and analyze voltage variations, contributing to the diagnosis of PQ.

In Figure 21, the waveform captured by the Hantek oscilloscope reveals the presence of harmonic distortion in the analyzed signal. A deformation of the sine wave is noticeable, especially at the wave peaks, which is characteristic

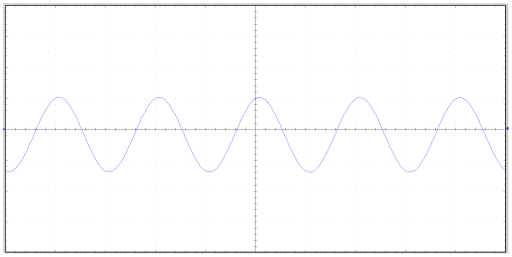


FIGURE 20. Voltage sag disturbance captured by the oscilloscope and identified by the prototype.

of the introduction of frequency components higher than the fundamental. This disturbance can be attributed to the presence of nonlinear loads in the electrical system, such as electronic devices, frequency converters, switched-mode power supplies, LED lamps, inverter-controlled motors, and other equipment that operate with semiconductor components.

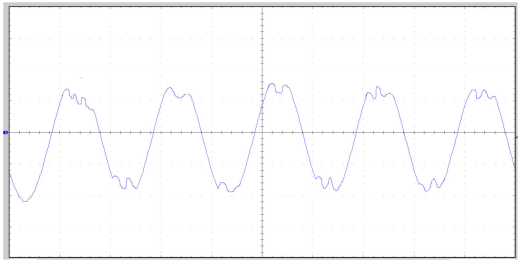


FIGURE 21. Harmonics present in crests and valleys, typically caused by non-linear loads.

Another type of disturbance addressed in this study is the oscillatory transient, characterized by high-frequency oscillations superimposed on the fundamental sinusoidal waveform, as illustrated in Figure 22. This type of transient typically occurs near the peak of the voltage waveform, suggesting that it may have been triggered by a switching operation or the energization of a substantial inductive load. The presence of high-frequency components at the crest of the waveform is indicative of a typical oscillatory transient, often associated with events such as capacitor bank energization, contactor disconnection, or resonance phenomena within the electrical network.

The observed transient appears to exhibit relatively fast damping, which suggests that the disturbance source was rapidly attenuated by the system's inherent impedance. This behavior is consistent with damped oscillatory responses that result from low-loss transient events, and it reinforces the importance of high-resolution signal acquisition for accurately capturing and classifying such phenomena.

The signal shown in Figure 23 contains a spike-type disturbance, which is characterized by short-duration, high-amplitude peaks superimposed on the sinusoidal waveform. The disturbance appears as abrupt and short-lived peaks in the waveform. These peaks are typically high-frequency and

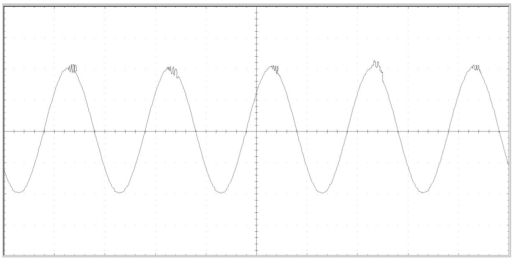


FIGURE 22. Oscillatory transient recorded on the oscilloscope, which can cause malfunctions in electronic systems and increase the wear of electrical insulation.

can be positive, negative, or both. Spikes are extremely short transients, differing from oscillatory transients by lacking noticeable damping.

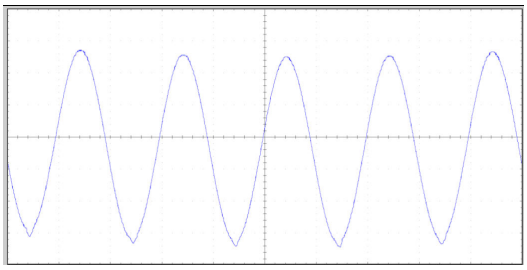


FIGURE 23. Spike-type disturbance, which can be mitigated with the use of a power line filter.

Finally, the signal shown in Figure 24 exhibits a notch-type disturbance, characterized by abrupt voltage drops throughout the waveform cycle. This phenomenon appears as notches in the sinusoidal wave, usually occurring at the zero crossing or other points in the cycle. It can be short in duration and may have a repetitive nature.

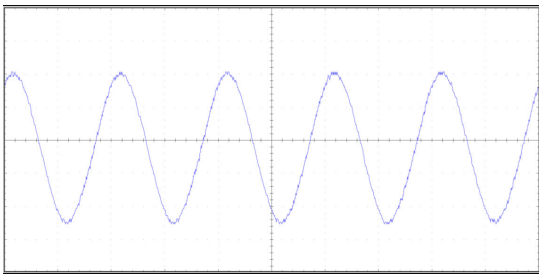


FIGURE 24. Notch-type disturbance, which can impair the operation of sensitive electronic equipment.

Although the prototype has demonstrated satisfactory performance in detecting voltage sags in accordance with the thresholds established by the PRODIST guidelines, certain limitations have been identified. One of the primary constraints is the system's current ability to detect only a single disturbance per acquisition window. This limitation may hinder the identification of consecutive or overlapping disturbances that occur in rapid succession, potentially compromising the completeness of PQ diagnostics.

Furthermore, the system's temporal resolution plays a critical role in the precise characterization of short-duration events. When compared to the Hantek 6022BE oscilloscope, which offers a significantly higher sampling rate, the prototype exhibits minor discrepancies in accurately determining the amplitude and duration of voltage sags. These deviations highlight the need for enhanced sampling strategies and optimized signal processing techniques to improve the fidelity of disturbance analysis in future iterations.

Another limitation of the system concerns its dependence on the quality and stability of the acquired electrical signals. Noise present in the electrical network or unexpected oscillations may compromise the accuracy of disturbance detection and classification. This underscores the need for improvements in the filtering stages and signal analysis algorithms to increase the system's robustness and resilience under adverse conditions.

Nevertheless, the results obtained demonstrate that the developed prototype holds promise for practical applications in PQ monitoring. Its ability to detect and classify disturbances in real time validates the core methodology. However, further enhancements are required—particularly in extending its capability to accurately detect and analyze multiple overlapping disturbances within a single acquisition window, thereby increasing its effectiveness in more complex and dynamic electrical environments.

Field deployments typically exhibit overlapping noise and multiple simultaneous events. To assess robustness, additional tests with injected Gaussian noise (SNR = 20 dB) showed detection accuracies above 85% for all disturbance types. Nonetheless, integrating adaptive filtering or wavelet-based denoising could further improve resilience to severe noise and overlapping disturbances, which we identify as a direction for future work.

In addition to the qualitative waveforms, a quantitative comparison was carried out between the prototype and

TABLE 6. Quantitative comparison of PQ disturbance parameters measured by the prototype and oscilloscope.

Disturbance Type	Parameter	Prototype	Oscilloscope
Voltage Sag	Depth (%)	18.7	18.5
	Duration (ms)	148	145
Voltage Swell	Depth (%)	+12.4	+12.2
	Duration (ms)	162	160
Harmonics	THD (%)	6.3	6.1
Oscillatory Transient	Peak (V)	311	315
	Duration (µs)	415	410
Spike	Peak (V)	248	253
	Duration (µs)	57	55
Notch	Depth (%)	7.9	7.6
	Duration (µs)	96	92

the oscilloscope to further strengthen the analysis. Table 6 summarizes key parameters measured during representative PQ events, including sag depth and duration, transient peak amplitude, and THD. The results demonstrate close agreement between the two instruments, with deviations consistently within acceptable margins. This quantitative validation reinforces the accuracy and reliability of the proposed system in practical monitoring scenarios.

VI. CONCLUSION

The developed low-cost system performed well in recording the electrical consumption of equipment when tested with both inductive and resistive loads. In terms of PQ, the system demonstrated high accuracy, particularly for Voltage Sag and Voltage Swell disturbances. For Spike disturbances, accuracy exceeded 90%, while for Oscillatory Transient and Notch disturbances, the values were slightly below 90%. This outcome is expected, as these disturbances are inherently more complex to analyze.

Beyond the engineering implementation, this work also contributes to advancing the scientific understanding of IoT-based smart metering. The results demonstrate that the integration of consumption monitoring with PQ disturbance detection in a single, low-cost device enables not only reliable energy tracking but also real-time diagnostic capabilities within residential environments. This dual functionality highlights a new perspective on IoT applications, where embedded systems evolve from being simple data collectors to active agents in detecting, classifying, and responding to electrical anomalies. Such findings strengthen the role of IoT as a key enabler of intelligent, resilient, and cost-effective energy management solutions, paving the way for broader adoption in smart homes and future grid modernization initiatives.

Although the obtained results are promising, there is room to expand the scope of the research. Incorporating machine learning algorithms could enable the development of predictive functionalities, such as anticipating consumption peaks or detecting failures in electrical equipment.

Another interesting avenue for future work is the economic feasibility analysis for large-scale production of the smart meter, considering cost-effectiveness and its potential long-term impact on energy savings. Additionally, exploring the system's integration with other home automation technologies would be valuable, expanding its application in smart home solutions.

REFERENCES

[1] J. Wang, M. K. Lim, C. Wang, and M.-L. Tseng, "The evolution of the Internet of Things (IoT) over the past 20 years," *Comput. Ind. Eng.*, vol. 155, May 2021, Art. no. 107174.

[2] D. Bandyopadhyay and J. Sen, "Internet of Things: Applications and challenges in technology and standardization," *Wireless Pers. Commun.*, vol. 58, no. 1, pp. 49–69, May 2011.

[3] P. Schavemaker and L. van der Sluis, *Electrical Power System Essentials*, 1st ed., Hoboken, NJ: Wiley, 2008.

[4] A. Noureddine, R. Rouvoy, and L. Seinturier, "A review of energy measurement approaches," *ACM SIGOPS Operating Syst. Rev.*, vol. 47, no. 3, pp. 42–49, Nov. 2013.

[5] F. Leferink, C. Keyer, and A. Melentjev, "Static energy meter errors caused by conducted electromagnetic interference," *IEEE Electromagn. Compat. Mag.*, vol. 5, no. 4, pp. 49–55, 4th Quart., 2016.

[6] H. A. Hseiki, A. M. El-Hajj, Y. O. Ajra, F. A. Hija, and A. M. Haidar, "A secure and resilient smart energy meter," *IEEE Access*, vol. 12, pp. 3114–3125, 2024.

[7] C. M. Colson and M. H. Nehrir, "A review of challenges to real-time power management of microgrids," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Jul. 2009, pp. 1–8.

[8] D. O. Koval, "Power system disturbance patterns," *IEEE Trans. Ind. Appl.*, vol. 26, no. 3, pp. 556–562, Aug. 1990, doi: 10.1109/28.55958.

[9] J. Lamoree, D. Mueller, P. Vinett, W. Jones, and M. Samotyj, "Voltage sag analysis case studies," *IEEE Trans. Ind. Appl.*, vol. 30, no. 4, pp. 1083–1089, Aug. 1994, doi: 10.1109/28.297926.

[10] M. Aredes and R. M. Fernandes, "A unified power quality conditioner with voltage SAG/SWELL compensation capability," in *Proc. Brazilian Power Electron. Conf.*, Sep. 2009, pp. 1–6.

[11] M. H. J. Bollen, E. Styvaktakis, and I. Y. H. Gu, "Categorization and analysis of power system transients," *IEEE Trans. Power Del.*, vol. 20, no. 3, pp. 2298–2306, Jul. 2005.

[12] A. A. Alkahtani, S. T. Y. Alfalahi, A. A. Athamneh, A. Q. Al-Shetwi, M. B. Mansor, M. A. Hannan, and V. G. Agelidis, "Power quality in microgrids including supraharmonics: Issues, standards, and mitigations," *IEEE Access*, vol. 8, pp. 127104–127122, 2020.

[13] A. Viswanath, K. J. Jose, N. Krishnan, S. S. Kumar, and K. P. Soman, "Spike detection of disturbed power signal using VMD," *Proc. Comput. Sci.*, vol. 46, pp. 1087–1094, Jan. 2015.

[14] J. Levin, N. O. Pérez-Arancibia, and P. A. Ioannou, "Adaptive notch filter using real-time parameter estimation," *IEEE Trans. Control Syst. Technol.*, vol. 19, no. 3, pp. 673–681, May 2011.

[15] I. S. Hudan and T. Rijanto, "Rancang bangun sistem monitoring Daya listrik pada kamar kos berbasis Internet of Things (IoT)," *Jurnal Elektronika dan Telekomunikasi*, vol. 8, no. 1, pp. 91–99, 2019.

[16] S. Muralidhara, N. Hegde, and P. Rekha, "An Internet of Things-based smart energy meter for monitoring device-level consumption of energy," *Comput. Electr. Eng.*, vol. 87, Oct. 2020, Art. no. 106772.

[17] N. Dahlan, A. A. M. Aris, M. Saidin, M. Nadzeri, M. Nawi, W. Abbas, A. F. Abidin, H. Mohammad, Z. Zakaria, and P. M. Arshad, "Development of web-based real-time energy monitoring system for campus university," *J. Telecommun.*, vol. 8, no. 10, pp. 157–164, 2016.

[18] F. Jonathan and T. Wahyu Oktaviana Putri, "Prototype of home power monitoring tool for electrical outlet using ESP32," in *Proc. Int. Conf. ICT Smart Soc. (ICISS)*, Jakarta, Indonesia, Aug. 2021, pp. 1–10.

[19] K. C. Okafor, G. C. Ononiwu, U. Precious, and A. C. Godis, "Development of Arduino based IoT metering system for on-demand energy monitoring," *Int. J. Mechatronics, Electr. Comput. Technol.*, vol. 7, no. 23, pp. 3208–3224, 2017.

[20] T. R. Lin, N. H. O. Khan, and M. Z. Daud, "Arduino based appliance monitoring system using sct-013 current and zmpt101b voltage sensors," *Int. J. Energy Monitor.*, vol. 48, no. 9, p. 19, 2021.

[21] A. B. A. Aziz, S. A. H. B. Amirrudin, and L. B. Raya, "Analysis and monitoring energy consumption in basic electric bills," in *Proc. 19th IEEE Int. Colloq. Signal Process. Its Appl. (CSPA)*, Mar. 2023, pp. 1–6.

[22] M. O. Onibonoje, N. I. Nwulu, and P. N. Bokoro, "An Internet-of-Things design approach to real-time monitoring and protection of a residential power system," in *Proc. IEEE 7th Int. Conf. Smart Energy Grid Eng. (SEGE)*, Aug. 2019, pp. 113–119.

[23] B. Su and L. Wang, "Application of proteus virtual system modelling (VSM) in teaching of microcontroller," in *Proc. Int. Conf. E-Health Netw. Digit. Ecosyst. Technol. (EDT)*, vol. 2, Apr. 2010, pp. 375–378.

[24] M. C. C. Leite, F. A. M. Vieira, V. B. Silva, M. Z. Fortes, and D. H. N. Dias, "Harmonic analysis of a photovoltaic system connected to low voltage grid," *IEEE Latin Amer. Trans.*, vol. 16, no. 1, pp. 112–117, Jan. 2018.

[25] R. Oliveira, B. P. Ferraz, R. Agustini, R. Ferraz, and R. C. Leborgne, "Comparative overview of voltage sags standards: IEEE 1564 versus Brazilian regulatory procedures," in *Proc. Simposio Brasileiro de Sistemas Eletricos (SBSE)*, May 2018, pp. 1–6.

[26] A. Heenkenda, A. Elsanabary, M. Seyedmahmoudian, S. Mekhilef, A. Stojcevskimn, and N. F. A. Aziz, "Unified power quality conditioners based different structural arrangements: A comprehensive review," *IEEE Access*, vol. 11, pp. 43435–43457, 2023.
[27] D. Razmi, T. Lu, B. Papari, E. Akbari, G. Fathi, and M. Ghadamyari, "An overview on power quality issues and control strategies for distribution networks with the presence of distributed generation resources," *IEEE Access*, vol. 11, pp. 10308–10325, 2023.
[28] M. Bajaj, A. K. Singh, M. Alowaidi, N. K. Sharma, S. K. Sharma, and S. Mishra, "Power quality assessment of distorted distribution networks incorporating renewable distributed generation systems based on the analytic hierarchy process," *IEEE Access*, vol. 8, pp. 145713–145737, 2020.
[29] N. Abas, S. Dilshad, A. Khalid, M. S. Saleem, and N. Khan, "Power quality improvement using dynamic voltage restorer," *IEEE Access*, vol. 8, pp. 164325–164339, 2020.
[30] R. Kaushik, O. P. Mahela, and P. K. Bhatt, "Hybrid algorithm for detection of events and power quality disturbances associated with distribution network in the presence of wind energy," in *Proc. Int. Conf. Adv. Comput. Innov. Technol. Eng. (ICACITE)*, Greater Noida, India, 2021, doi: 10.1109/ICACITE51222.2021.9404665.
[31] A. Manjula, U. T. Kute, C. V. K. Reddy, and B. Mallala, "Power quality improvement of microgrid for photovoltaic ev charging station with hybrid energy storage system using RPO-ADGAN approach," *J. Energy Storage*, vol. 108, Feb. 2025, Art. no. 114970.
[32] B. Mallala, P. V. N. Prasad, and K. Palle, "Power quality conditioner with fuzzy logic controller," in *Proc. Inf. Commun. Technol. Competitive Strategies (ICTCS)*, May 2023, pp. 627–636.
[33] B. Vivek, B. H. Teja, B. Mallala, and G. Srinitha, "Electrical fault detection and localization using machine learning," in *Proc. Int. Conf. Expert Clouds Appl. (ICOECA)*, Apr. 2024, pp. 820–825.
[34] B. Mallala, P. A. Khan, B. Pattepu, and P. R. Eega, "Integrated energy management and load forecasting using machine learning," in *Proc. 2nd Int. Conf. Sustain. Comput. Smart Syst. (ICSCSS)*, Coimbatore, India, Jul. 2024, pp. 1004–1009.



THALLYS B. DE SOUSA received the degree in electrical engineering from the Federal University of Piauí (UFPI), in 2021, where he is currently pursuing the master's degree in electrical power systems. He also holds a specialization in environmental engineering and renewable energies. Throughout his academic and professional journey, he has developed skills in applied electronics, electrical circuit simulation, 3D modeling, and 3D printing. His technical profile focuses on the integration of sustainability and technological innovation within the electrical sector.



RICARDO DE A. L. RABELO received the B.Sc. degree in computer science from the Federal University of Piauí, Brazil, in 2005, and the Ph.D. degree in power systems from São Carlos Engineering School, University of São Paulo, Brazil, in 2010. He is currently a Professor with UFPI. He has authored or co-authored over 150 papers in refereed international journals and conferences. His research interests include smart grids, the Internet of Things, intelligent systems (computational intelligence), and optimization (linear and nonlinear programming).

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - ROR identifier: 00x0ma614